



With the advent of the internet, the very concept of freedom is something that has undergone significant change. Concepts such as open source software, and copy-left licences are certainly no invention of the internet, yet without it, their very model becomes infeasible. Today, most popular web sites, be it Wikipedia, Flickr, YouTube or even Twitter, are populated with user-generated content.

Technology advances freedom and nurtures it. Your computer alone can be the source of information, entertainment, and even income. Yet, we are still constrained by the physical world we live in, and even the best internet connections cannot assuage a hostile government.

In August, this year, we celebrated the 62nd anniversary of our Independence Day. To commemorate this occasion, we decided to compile a list of some of India's youngest innovators. Some of India's brightest and youngest minds, who have unveiled innovations that are bound to influence the shape of things to come.

Our goal was to introduce to you (our readers), some of

the brightest minds from this country and to garner your opinion on the most influential innovation. As such, we conducted a poll at <http://www.thinkdigit.com/yii>, allowing you to browse through profiles, and vote for your most deserving candidate.

No innovation is insignificant; in today's world, the inventor of the socket is as responsible for our computers switching on, as the one who discovered the transistor. The smallest of innovations pile up today to create tomorrow's world.

Here are some of the participants:

Anand Agarawala

As operating systems progress, we tend to see a move towards interfaces that emulate the real world. Anand's software BumpTop takes it several steps forward in this regard, and turns your desktop into a realistic 3D physical-desktop-like environment.

We are used to interpreting the reality around us as physical objects, and by displaying objects inside your computer. BumpTop removes extraneous layers of interpretation. Objects on your BumpTop desktop obey similar rules of placement, and interaction, by allowing



Anand Agarawala

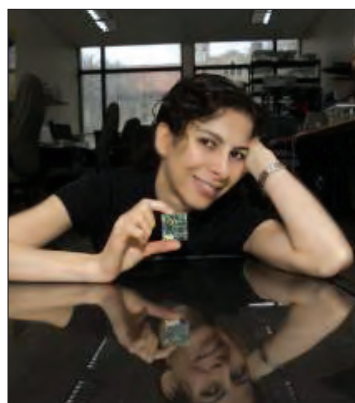
stacking of objects, and tossing documents to a printer, or USB drive; commonly used files can even grow in size, making them easier to find.

Agarawala's application, created as a Masters' project, has even been presented at TED Talks and has significantly evolved ever since.

Farinaz Koushanfar

With rampant piracy brewing around us, it's quite easy to duplicate software – with or without the internet. Yet, hardware piracy, although a lesser-known phenomenon, can be perhaps much more damaging. ICs could be used anywhere from innocent video players to high grade weapons.

Koushanfar's technique uses a unique ID to lock each chip so that it cannot be used without either a redesign, or a code from the designer of the circuit, making hardware piracy a tough nut to crack. Her method uses inherent manufacturing imperfections, as a base for creating the



Farinaz Koushanfar

unique code. Tiny deviations in the circuit, which have no effect on the device otherwise are encoded as a binary sequence (1s and 0s) and makes each IC unique.

Priti Ranadive

Even as the number of computing cores increase, software applications are slow to catch up. With dual and quad-core computers becoming commonplace, a lot of thought and effort needs to be put into designing a software to take full advantage of the performance increases which come with parallelism.

Ranadive's tool enables software to keep up with the multi-core revolution. This automatic parallelization tool helps convert legacy source code to parallel code and exploit the advantages offered



Priti Ranadive

by multi-cores. Arguably, this tool's best feature is that it requires little or no manual effort, and stands to offer great performance gains and reduce the execution time of the code. This can be successfully implemented in the field of bioinformatics in applications such as protein folding, weather prediction, mathematical simulations among others. Such fields stand to gain a lot by being optimised for parallel processing.

The young Indian innovator with the most innovative and influential creation is...

Pranav Mistry

This young innovator has already garnered much fame and attention around the world